



Can ASR generate valid measures of child reading fluency?

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Abstract

Early diagnosis of reading difficulties requires assessment of children's oral reading fluency. Since obtaining reliable assessments through subjective rating rubrics has proven to be difficult and time consuming, the current study presents a novel procedure incorporating Automatic Speech Recognition (ASR) to compute 15 measures related to the aspects phrasing, smoothness and pacing of oral reading fluency. We investigate the validity of these measures by comparing them to the same measures computed on human transcripts. This comparison was performed on a dataset of 244 recordings of texts read by 131 Dutch primary school children (aged 6-13 years). We found strong correlations for 12 out of 15 measures, which emphasizes the great potential of these measures for more reliable, less time-consuming and sustainable reading assessment.

Index Terms: speech recognition, reading fluency, children, automatic assessment, validity

1. Introduction

The ability to read written text fluently is one of the core skills children acquire in primary school. Fluent reading is defined as the combination of accuracy, automaticity and oral reading prosody to facilitate the reader's construction of meaning [1]. Several tests have been developed to measure oral reading proficiency (e.g. *AVI* [2] in Dutch and *DIBELS* [3] for English). Although widely used, these tests have several limitations. They are time-consuming, they rely on the subjective assessment of the teacher (word is read correctly or not), and they provide only shallow reading diagnostics by returning just one measure, *words correct per minute (wcpm)*, which describes the number of words read correctly per minute. To obtain a good picture of reading proficiency, it is essential to develop assessment instruments that extend beyond measures of automatic word recognition, for example by incorporating measures of oral reading prosody [4], which is an umbrella term for the phenomena of stress, rhythm, phrasing and intonation [5].

Rating rubrics have been developed that also include measures of oral reading prosody, such as the Comprehensive Oral Reading Fluency Scale (CORFS) [6], National Assessment of Educational Progress (NAEP) fluency scale [7] and Multidimensional Fluency Scale (MDFS) [8, 9]. In these rubrics, oral reading fluency is scored on three (CORFS: wcpm, intonation, pausing), one (NAEP), and four (MDFS: expression & volume, phrasing, smoothness, pacing) different scales, and the scores on the different scales are summed to obtain one overall score of reading fluency. This summation is performed because the scores on the individual scales do not have sufficient reliability. Despite summing, obtaining reliable scores remains difficult and laborious, since raters need to be trained and have to

rate multiple passages [10]. This means that using these rubrics for reading fluency assessments in classrooms is not feasible.

One way of reducing the assessment load for teachers and obtaining more objective and advanced reading diagnostics would be by employing Automatic Speech Recognition (ASR) technology to automatically transcribe read speech and align this transcription with the speech signal and with the prompt. In this way, measures related to accuracy and automaticity can be extracted, and parameters underlying oral reading prosody (e.g., duration, amplitude and fundamental frequency (F0)) can be measured at text, sentence and word level through spectrographic analysis. A fully automatic pipeline can thus be developed to obtain measures related to oral reading fluency from speech, which are reliable since they can be easily reproduced.

Previous studies have shown that measures extracted from read speech through an automatic ASR-based approach can predict subjective oral reading fluency scores of child read speech on a 3-level [11] or 6-level [12] ordinal scale, and on a rating rubric scale, such as NAEP [13] or MDFS [14]. These studies reported the ASR transcription quality, but did not investigate the validity of ASR-based oral reading fluency measures. Although speech recognition performance has drastically improved in recent years, obtaining good transcription accuracy for child speech is still challenging [15, 16, 17]. On the other hand, it is possible that even moderate performance is sufficient to compute valid automatic ASR-based measures from read child speech. To gain more insights into the potential of ASR-based reading proficiency scoring, we conducted research aimed at validating automatic ASR-based measures by comparing them to measures computed from time-aligned human transcriptions. In this paper we report on a study that addressed the following research question: *To what extent can ASR generate valid measures of oral reading fluency for primary school children?*

2. Methodology

2.1. Speech datasets

We used three different datasets of Dutch child read speech: JASMIN-NL, JASMIN-VL and SERDA-NL. An overview of these datasets is presented in Table 1. JASMIN-NL and JASMIN-VL were extracted from comp-q of the open source JASMIN-CGN Corpus¹ [18]. JASMIN-NL consisted of 142 recordings by native Dutch child speakers, JASMIN-VL contained 84 recordings by native Flemish child speakers. Each speaker read two different texts at their own level, which were saved as two separate recordings. There were nine reading lev-

¹Version 1.0 (2008). Available at the Dutch Language Institute: <http://hdl.handle.net/10032/tm-a2-j7>

els defined in the original corpus, so a total of 18 different prompts (i.e., texts) were read in both JASMIN datasets. The third dataset that was used is SERDA-NL. This corpus is a selection of 18 recordings from the SERDA corpus [19]. The speakers were Dutch children from grade 2 and 3, who read in each recording a text at, above or below their reading level. There were three reading levels defined in the original corpus, so a total of three different prompts (i.e., texts) were read in the SERDA dataset. Time-aligned orthographic transcriptions were available in TextGrid format [20] for all three datasets [18, 21]. The final dataset consisted of 244 recordings of read texts by primary school children and contained maximally two recordings of the same speaker. Each age group was well presented in the dataset: 34% of the recordings contained speech by children from 6-8 years old, 35% were of children from 9-10 years old, 26% were of children from 11-13 years old and 5% were from children for whom the age is unknown.

Table 1: Overview of child read speech datasets.

Data set	Recordings		Speakers	
	<i>N</i>	<i>Mean dur.</i> (mm:ss)	<i>N</i>	<i>Age range</i> (years)
JASMIN-NL	142	03:35	71	6-13
JASMIN-VL	84	02:54	42	7-12
SERDA-NL	18	02:03	18	7-9
Final dataset	244	03:14	131	6-13

2.2. Selection of oral reading fluency measures

We created a set of 15 objective measures related to oral reading fluency (see Table 2). We based this set on reader characteristics related to phrasing, smoothness and pacing that are described in MDFS. The selected measures are those that could be validated by comparing them to manual transcriptions. We included 9 measures related to phrasing, which are **all_paurate**, **ia_paurate**, **ir_paurate**, **all_paudur_mean**, **ia_paudur_mean**, **ir_paudur_mean**, **all_paudur_std**, **ia_paudur_std**, and **ir_paudur_std**. These measures describe all (all), intersentential (ir) and intrasentential (ia) pauses by computing for each type of pauses the pause rate, mean duration and std duration. These phrasing measures are assumed to capture children's sense of phrasing, which according to the MDFS, ranges from little sense of phrase boundaries, to two and three word phrases, to mid-sentence pauses for breath, to well-phrased in clause and sentence units. Secondly, we defined 4 measures related to smoothness, which are the number of words read correctly per minute (**wcpm**), the percentage of prompt words that are not read and thus deleted (**perc_del**), the number of non-prompt words that are inserted (e.g., false-starts, repetitions and multiple attempts) divided by the number of prompt words (**perc_ins**), and the percentage of prompt words that are read incorrectly, and thus substituted by another word (**perc_sub**). In the MDFS, smoothness ranges from reading with frequent hesitations and repetitions to smooth reading in which the child self-corrects the reading of difficult words. Thirdly, we defined 2 measures related to pacing, which are speech rate **sp_rate** as the number of words read per minute audio, and articulation rate (**art_rate**) as the number of words read per minute speech (voiced audio). In the MDFS, pacing ranges from reading slowly, to an uneven mixture of fast and slow reading, to reading consistently at a conversational pace.

Table 2: Overview of the selected fluency measures.

Feature	Tag	Unit
Phrasing		
intersentential pauses		
mean duration	ir_paudur_mean	s
std duration	ir_paudur_std	s
rate	ir_paurate	#ir pauses/min
intrasentential pauses		
mean duration	ia_paudur_mean	s
std duration	ia_paudur_std	s
rate	ia_paurate	#ia pauses/min
all pauses		
mean duration	all_paudur_mean	s
std duration	all_paudur_std	s
rate	all_paurate	#all pauses/min
Smoothness		
words correct per min.	wcpm	#cor/min
% of substituted words	perc_sub	#sub/#prompts
% of deleted words	perc_del	#del/#prompts
% of inserted words	perc_ins	#ins/#prompts
Pacing		
speech rate	sp_rate	#wds/min
articulation rate	art_rate	#wds/min speech

2.3. Computation of oral reading fluency measures

To compute these measures automatically from speech, we developed a pipeline that consists of two modules: an ASR module followed by an alignment module. The code is openly available². In the ASR module, the time-aligned transcriptions of the recordings were obtained using the Dutch Large-v2 ASR model from Whisper Timestamped [22]. This model extends the Whisper model [23] by using Dynamic Time Warping (DTW) [24] to compute word-level timestamps. We generated for each recording 4 different transcriptions by using different input settings of the ASR model: *whisper* uses the default settings, *whisper_dis* has enabled Disfluency Detection (DIS), *whisper_vad_dis* has enabled DIS and Voice Activity Detection (VAD), and *whisper_prompt* has the initial prompt property set to the prompt. The ASR output with DIS enabled contains intervals with the tag *[*]*, which represent disfluencies. Through informal listening, we observed that these tags mainly mark respiratory breaths, but also background noise, filled and silent pauses, false starts and incorrect readings. The silent pauses occur most frequent, and for that reason we decided to treat the disfluency intervals as pauses. From the time-aligned ASR transcriptions, we computed 5 fluency measures: the two pacing measures and the three phrasing measures that are related to all pauses. For validation purposes, the same 5 measures were also obtained directly from the orthographic transcriptions.

Using the alignment module, the other 10 fluency measures are obtained from both the time-aligned ASR transcriptions and time-aligned orthographic transcriptions. The same procedure is followed for both transcriptions. In the alignment module, each transcription is aligned with its prompt. This alignment provides insights on how specific words and sentences are read by the children. A difficulty in this procedure is that children can omit reading a sentence. The proposed procedure takes this into account. First the prompt and transcript are

²<https://github.com/WiekeHarmsen/readingFluencyFeatureSet>

aligned globally using the PairwiseAligner of the Biopython³ package [25]. Next, the segments where one word is aligned with multiple words are selected, and these are a second time aligned, this time using word alignment (implemented using the Needleman-Wunsch algorithm from the Python package string2string⁴ [26]). Now each word in the prompt is aligned with maximally one word from the transcript, which makes it possible to map the word timestamps of the transcripts with each prompt-transcript word alignment. Needleman-Wunsch is used again to align the characters of the aligned words. In this resulting alignment, we can compare each prompt word with its transcription and obtain the 4 measures related to smoothness. In addition, we can classify pauses as intersentential or intrasentential in the resulting alignment, which allows for the computation of the 6 phrasing measures related to intersentential and intrasentential pauses. For the computation of all 9 phrasing measures, we only included pauses longer than 0.2s.

2.4. Validation of ASR-based oral reading fluency measures

We first determined which of the four ASR systems provided the most accurate speech transcriptions. To this end, we evaluated the transcription accuracy of the four systems by computing Word Error Rate (WER), Substitution Error Rate (SER), Insertion Error Rate (IER) and Deletion Error Rate (DER) using the tool SCLITE from the NIST SCoring ToolKit (SCTK)⁵. We normalized both ASR and orthographic transcriptions first by converting them to lowercase, removing punctuation, writing numbers as text⁶, normalizing the spelling of names, and correcting the most frequent spelling errors. In addition, we also evaluated both the transcription and timing accuracy by computing precision, recall and the F1-score in line with Bain et al. (2023) [27] and Zusag et al. (2024) [28]: $Precision = TP/N_{ASR}$ and $Recall = TP/N_{ORT}$ where N_{ASR} is the number of recognized words by the ASR, N_{ORT} is the number of actually read words according to the orthographic transcription, and TP is the number of True Positives. A True Positive is a recognized word that has both an exact string match and temporal overlap with a reference word. Temporal overlap means that the start and end timestamps of a recognized word fall within a predefined collar of the corresponding reference word timestamps (see the light blue segments in Figure 1). We used a collar value of 0.2 s, since this is in line with previous studies [27, 28].

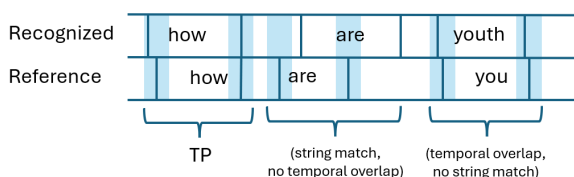


Figure 1: Example of TP for the evaluation of timing accuracy.

Subsequently, we validated the ASR-based reading fluency measures of the highest performing ASR system by computing Pearson’s correlation coefficient (using the Python package SciPy [29]) between the automatic measures computed from the ASR transcript and the human measures computed from orthographic transcriptions.

³<https://biopython.org>

⁴<https://github.com/stanfordnlp/string2string>

⁵<https://github.com/usnistgov/SCTK>

⁶<https://pypi.org/project/num2words/>

3. Results

3.1. Evaluating ASR transcription and timing accuracy

Table 3 shows the transcription accuracy (WER, SER, DER and IER) and transcription and timing accuracy (F1) of the four different ASR systems. With respect to transcription accuracy, we found comparable performance for the ASR systems *whisper*, *whisper_dis* and *whisper_vad_dis*, where results for *whisper_vad_dis* are slightly better for WER and SER. For these three systems, deletion errors are most frequent, which means that words present in the speech signal are missing in the ASR transcription. Inspection of these deletion errors showed that most of them are self-corrections or repetitions (e.g., *bekijkt bekijkt* is transcribed as *bekijkt* (looks at), the incorrect first attempt *bekijkt* is not transcribed.). In addition, the most frequent substitution error is the ASR transcribing the indefinite article *een* (a) as the definite article *de* (the).

When evaluating the ASR systems on both transcription and timing accuracy, we found that both *whisper_dis* and *whisper_vad_dis* show good performance ($F1 \geq 0.8$), and that *whisper_vad_dis* shows the best performance (see Table 3). In the context of fair and inclusive speech science, we investigated whether this finding holds for all age groups and visualized the distribution of F1-scores of *whisper_vad_dis* for each age group in the dataset (see Figure 2). We found for all age groups a mean F1-score of 0.8 or higher, except for the seven-year-olds ($F1_{mean} = 0.67$). In this age group, the F1-scores also show a high standard deviation ($std = 0.18$) in comparison to the other age groups. In addition, we observe that F1 increases over the years till the score stabilizes around a value of 0.9 from age = 10. The group of six-year-olds are an outlier on this trend, which can possibly be explained by the fact that this group only consists of one student reading two texts.

Table 3: Evaluation of transcription and timing accuracy ($N=89,518$ reference words from 244 recordings). ‘wht’ is an abbreviation for ‘whisper’. The reported 95%-confidence intervals (ci) are computed using a binominal model.

ASR	WER (% ± ci)	SER (%)	DER (%)	IER (%)	F1
wht	12.4 ± 0.22	4.8	6.3	1.3	0.74
wht_dis	13.0 ± 0.22	4.8	6.3	1.9	0.80
wht_vad_dis	12.3 ± 0.22	4.5	6.3	1.5	0.82
wht_prompt	17.5 ± 0.25	8.7	5.5	3.4	0.63

3.2. Validity of ASR-based reading fluency measures

Since *whisper_vad_dis* performs best with respect to both transcription and timing accuracy, we compared the ASR-based reading fluency measures computed using this ASR system with the measures computed from human orthographic transcriptions. Table 4 shows Pearson’s correlation for each measure. For 12 of the 15 measures, we found strong correlations ($r \geq 0.7$). These are all pacing measures, the smoothness measures *wcpm* and *del_perc*, the phrasing measures related to intersentential and intrasentential pauses, and two measures related to all pauses (i.e., *all_paudur_mean*, *all_paudur_std*). We found moderate correlations ($0.5 \leq r < 0.7$) for the phrasing measure *all_paurate*, and the smoothness measure *sub_perc*. The smoothness measure *ins_perc* had a weak correlation ($r < 0.5$).

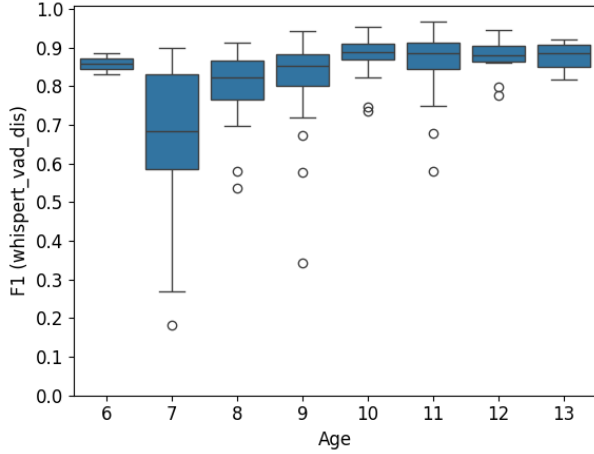


Figure 2: Boxplots representing the distribution of whisper_vad_dis F1-scores for each age group.

4. Discussion & Conclusion

In the current study, we investigated to what extent ASR can generate valid measures of oral reading fluency for primary school children. To answer this question we compared 15 ASR-based measures obtained automatically from three different corpora of child read speech to measures for the same recordings computed from time-aligned human transcriptions. Although the transcription and timing accuracy scores produced by the ASR system were not perfect (WER = 12.3%, F1 = 0.82), we found strong correlations ($r \geq 0.7$) for 12 out of 15 ASR-based measures related to pacing, phrasing and smoothness. The headmost interpretation of these results is that these 12 ASR-based measures provide valid indicators of oral reading fluency. However, the specific use of such measures needs to be considered with care depending on the context in which the measures are employed. For instance, for use in high-stakes tests, it is argued that valid measures require correlations of at least 0.9 [30, 31]. This means that among the measures we studied only four managed to satisfy this criterion. For other types of use in the educational context such as pre-screening or formative assessment other criteria may hold, which implies that possibly more measures can be usefully employed. In any case, these results provide evidence that this is a promising research direction, especially given the fact that this study concerns child speech, which is challenging to transcribe [15, 16, 17].

In addition, it is important to note that there is considerable room for improvement in the computation of the three ASR-based measures in which we did not find strong correlations. Two of these measures are related to smoothness, sub_perc (the percentage of prompt words containing a substitution error) and ins_perc (the number of inserted words relative to the prompt), whose low correlations we expect to be consequences of the relatively high substitution error rate (SER=4.5%) and deletion error rate (DER=6.3%), respectively. We also assume that this relatively high deletion error rate might explain the moderate correlation we found for the measure all_paurate (total number of pauses per minute), which is related to phrasing, since inspection of ASR deletion errors showed that repetitions and self-corrections are often not transcribed by the ASR and are therefore recognized as pauses, which increases the number of pauses recognized through ASR. To improve the validity of

Table 4: The measures sorted on correlation strength between manual and automatic (whisper_vad_dis) measures. Pearson’s correlation is presented between brackets.

	phrasing	smoothness	pacing
strong	ir_paurate (.89) ir_paudur_mean (.89) all_paudur_mean (.85) ia_paudur_mean (.83) ia_paudur_std (.82) all_paudur_std (.82) ir_paudur_std (.81) ia_paurate (.76)	wcpm (.99) del_perc (.98)	sp_rate (.98) art_rate (.93)
moderate	all_paurate (.51)	sub_perc (.67)	
weak		ins_perc (.32)	

these measures, it might be beneficial to improve the interpretation of the disfluency tag [*] by training a model to classify them as either respiratory breath, false start or incorrect reading. Another way of improving these results is by using another ASR model to transcribe the speech. Especially fine-tuning the Whisper model on child speech or on the specific texts that the children read, might be a promising research direction, since this has already shown good results for child speech in other languages [32]. The presented analysis pipeline has a modular architecture, which enhances flexibility and enables easy integration of new ASR models.

Another limitation is that the current work does not include the computation of prosodic measures, which are related to the MDFS scales phrasing and expression & volume. However, the current computation procedure implements the functionality to align the prompt, audio signal and transcript. This alignment reveals information on how specific words and sentences are read and paves the way for the computation of prosodic measures by measuring their underlying parameters duration, amplitude and F0 at word, sentence and text level. However, just computing these parameters is not sufficient to obtain informative measures of oral reading prosody. For this purpose, reference prosody patterns are essential, for example through prosodic prominence annotations in the prompt [33], or through comparison with adult reference speakers [14].

The current results seem to indicate that it is already possible to use these automatically calculated oral reading fluency measures to the benefit of reading education and assessment. On a global scale, such measures could be used to identify specific reading profiles and to track the development of reading skills in individual children and/or groups of children in classes. These data can become a rich resource for research that provides new insights into reading development. If incorporated in an interactive reading tutor with which children can practice reading aloud, these measures can be obtained from the read speech the children produce during practice, under realistic circumstances, in a sustainable way. In this way, possible reading difficulties can be detected in time and children can receive additional support before it is too late. Diagnostic information based on these measures can also be provided to teachers and parents, so that teaching can be adapted and possibly personalized to address individual problems.

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